

may restrict you from using the latest lead-less parts that have a pin pitch of 0.5mm or 0.4mm. If you need to cut a hole of 0.25mm x 0.25mm the typical desktop cutters face the limit of their capability here. For steel stencils you need to use ultraviolet lasers that are very expensive. You only find these at big contract manufacturers or you can get a custom stencil by sending your gerber to China from where you will get a steel stencil for about US\$ 80. Only do this for super-fine pitch parts with, say, 0.5mm pin pitch.

You may also want to go for steel if you want to do volume as vellum will not stand up to high volumes. You can get a steel stencil framed with an aluminium frame that is designed for mounting into the stencil printer.

You might be able to order a stencil at US\$ 80 in China, but to get that to India you will end up paying another US\$ 200. Venu suggests you try the many local companies in India that make quality stencils.

When you are dealing with 0.25mm pitch holes, getting the stencil aligned to the board can be tricky business. Stencil printers can be used here. While manual ones are not really useful, automatic ones are very expensive.

Reflow ovens

Reflows are generally categorised in two ways.

Batch reflow. This happens in batches of PCBs.

Continuous reflow. With the help of a conveyor, it moves through the oven, all controlled by PID/PLC control with multi-zones of temperature for better reflow soldering. There are a few Chinese vendors that have very good desktop models of this type.

There are a number of reflow ovens in the market from reputed Chinese suppliers. There are some small reflow ovens that can handle up to 10mm² and larger ones that can go up to 330mm by 220mm, which is almost as big as an A4-size paper,

and that is a good size to work with for office production.

When it comes to reflow ovens, get the biggest one that you can afford. When you get bigger ones, you require good ventilation as these generate fumes and smell.

In the case of smaller desktop ones, you do not need to worry about fumes and smell under normal conditions. But if you are using these throughout the day, then you do need to worry about the fumes.

You are either done or you need to rework

No matter how good a machine is, you always have to inspect what the machine did for you and how the reflow went. Coggeshall says, "The biggest challenge that I have is the paste application and whether the paste is getting on in a uniform manner, or whether you have solder bridges due to too much or too little paste."

Other cases are where there are big inductors, which could take a long time to heat up due to their thermal mass and the profile of the board. This could cause you to not get complete reflow. So when the boards come out, you need to check these out in a rework station.

Nitish Gupta, business development, Advance Tech Services, says, "Defective surface-mount components can be repaired by using soldering iron or by using a non-contact rework system. In most cases, a rework system is the better choice because SMD work with a soldering iron requires considerable skill and is not always feasible. Temperature accuracy is of utmost concern. RX-802 has a patent six-seconds heat recovery used with a reflow bit for SMT soldering that delivers better and efficient working without shorting and bridging."

Coggeshall gets a piece of equipment called 898 rework station for

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about US\$ 88 through *AliExpress*. It is called so because the first Chinese company used that name and now eventually there are a dozen companies using the same name. Coincidentally, the ovens are the same, too. These all look the same on *AliExpress* or *Alibaba*. The 898 is a combination of hot air and soldering iron—the hot air is truly indispensable for surface mount rework. There is a set of skills you need to hone if you have to reflow, and you have to use a solder wick to remove solder.

Hot air will usually help you take care of most reflow unless you have something exotic or very small. An exotic component is a lead-less component like the one on Arduino Leonardo quad-flat no-leads where the central processing unit is a small black rectangle. If you have one in hand, look carefully at the perimeter and you will see small dots. That part is impossible to work on without hot air.

There is one more piece of equipment that is basically a bed that heats up—Aoyues66, a toaster oven that sits on your desk and costs approximately US\$ 200. When you place your board on the bed, it will heat up the whole board from the bottom. It also has a little arm over the board and a hot-air gun that blows hot air down on the board. These are the last resort if you have rework to do like when you have a ball grid array with pins on the bottom; this may be the only way to get to that part of the board. For light emitting diode arrays that are closely placed, this is the only way to get densely packed parts of the same. ●